

JA

Joshua Alcobia Gomes

Mechanical Engineer

Second-year MEng Mechanical Engineering student at UCL (predicted First), heading to McLaren on an industrial placement for 2026–27. I work across automotive and aerospace — from CFD and rocketry to pharmaceutical modelling, machine learning and AI tooling.

DISCIPLINE

Mechanical Engineering

UNIVERSITY

UCL · MEng

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LINKEDIN

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Curiosity, engineered.

I'm a mechanical engineer drawn to the fastest, most demanding corners of the field — motorsport and flight. I'm in my second year of an MEng in Mechanical Engineering at UCL (predicted First, 83%), and I'll be joining McLaren on an industrial placement for the 2026–27 academic year.

My work spans design, simulation, testing and the reality of making things perform: CFD in ANSYS Fluent, structural and generative design in Fusion 360 and SolidWorks, and machine-learning and AI tooling built during my scientific-modelling internship at Novartis. I lead teams as happily as I run the numbers — from an 8-engineer national rocketry squad to award-winning design challenges.

Away from engineering I compete nationally in track and field, which is where a lot of the discipline behind this work comes from — and the calm, natural palette of this site reflects where I recharge.

Fusion 360 · SolidWorks · Onshape

ANSYS Fluent & Mechanical

COMSOL · OpenRocket · QBlade

Python · MATLAB · C++

CFD & FEA

Machine Learning · LLMs

3D printing · laser cutting

AT A GLANCE

Discipline: MEng Mechanical Engineering**University:** UCL · 2nd year (predicted 1st, 83%)**Next:** McLaren — Industrial Placement 2026–27**Focus:** Automotive & Aerospace**Based in:** London, UK**Email:** jg7alcobia@gmail.com

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A separate 2-page résumé accompanies this portfolio.

McLaren

High-performance automotive engineering — my industrial placement year.

ROLE	TEAM	PERIOD	DOMAIN
Industrial Placement	TBC	2026–27 academic year	Automotive · Motorsport

I'll be joining McLaren for my industrial placement across the 2026–27 academic year — a full year embedded in one of the sharpest engineering environments in the world. This page will grow into a record of the projects I work on there.

APPROACH

1. Bring rigorous simulation and design fundamentals from UCL and NING Research.
2. Learn the pace and precision of elite motorsport engineering.
3. Document each project here as the placement progresses.

PROJECTS

3 projects

COMING SOON

Placement project 1

Project details will appear here once my placement begins in 2026.

COMING SOON

Placement project 2

Project details will appear here once my placement begins in 2026.

COMING SOON

Placement project 3

Project details will appear here once my placement begins in 2026.

NING Research

Applied CFD and engineering simulation — turning real client problems into validated models.

ROLE**Simulation / CFD Engineer****FIELD****CFD & Engineering Simulation****PERIOD****Start – present****DOMAIN****R&D · Consulting**

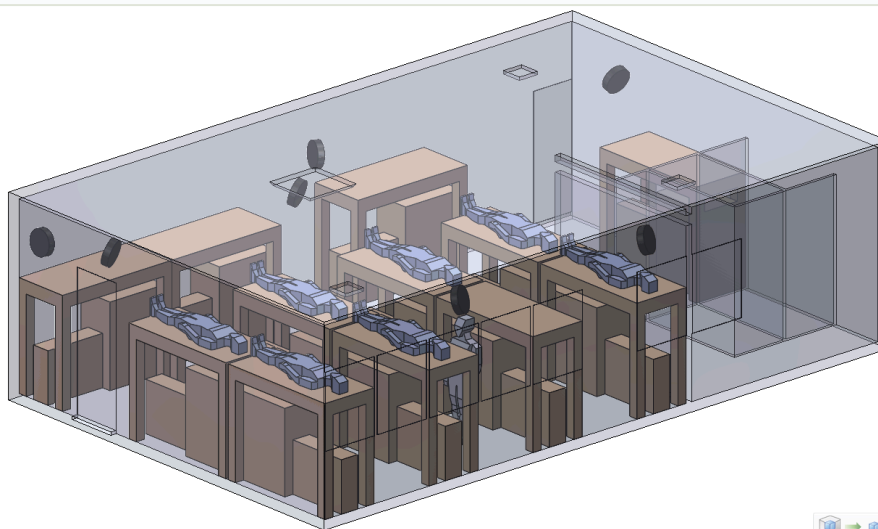
At NING Research I work on computational fluid dynamics and engineering-simulation projects for a range of clients — from ventilation and mixing to structural and environmental analysis.

APPROACH

1. Define the physical problem and success criteria with the client.
2. Build and mesh the CFD / simulation model (OpenFOAM, ANSYS).
3. Validate against data, then translate results into clear recommendations.

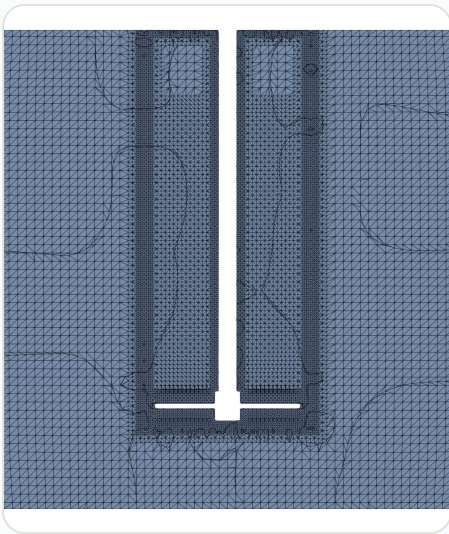
PROJECTS

9 projects

**CFD · VENTILATION**

Coughing Dispersal Simulation

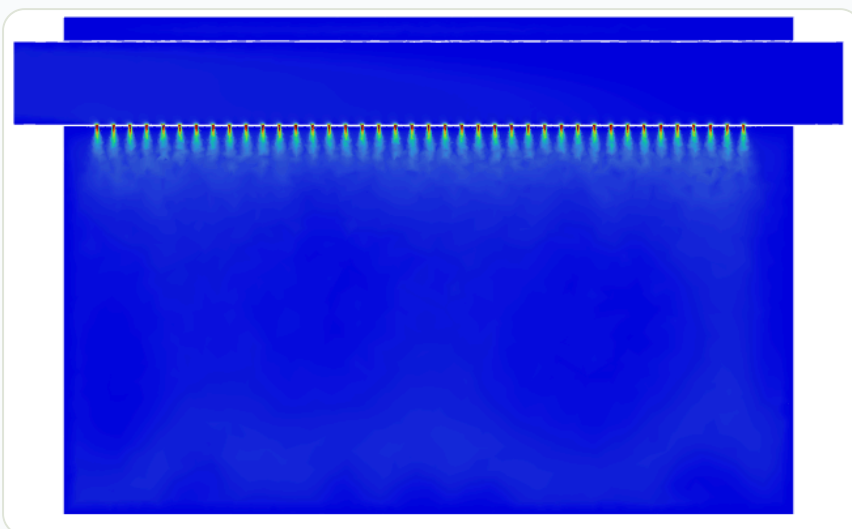
Full write-up in progress — imagery above. Industry: healthcare / built-environment ventilation.



CFD · MIXING

OpenFOAM Mixing Tank

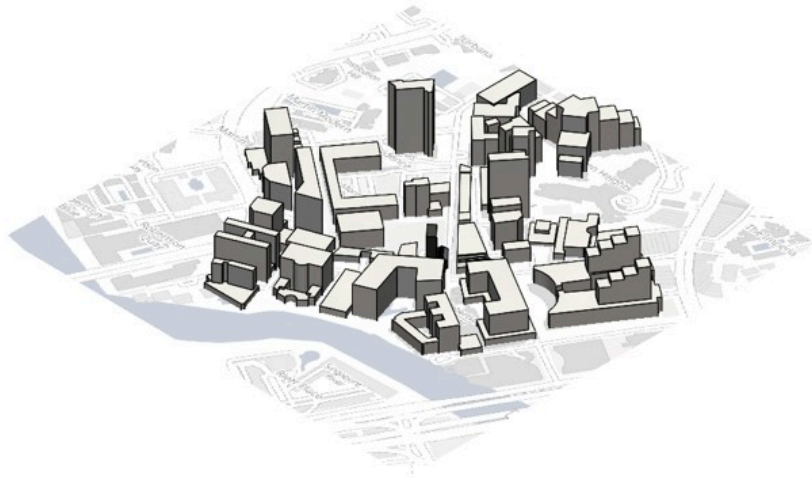
Full write-up in progress — imagery above. Industry: industrial process engineering.



SIMULATION

District-Scale Simulation Study

Full write-up in progress — imagery above. Industry: urban / built-environment simulation, Singapore.



CFD · BUILT ENVIRONMENT

Mixed-Use Development Ventilation Study

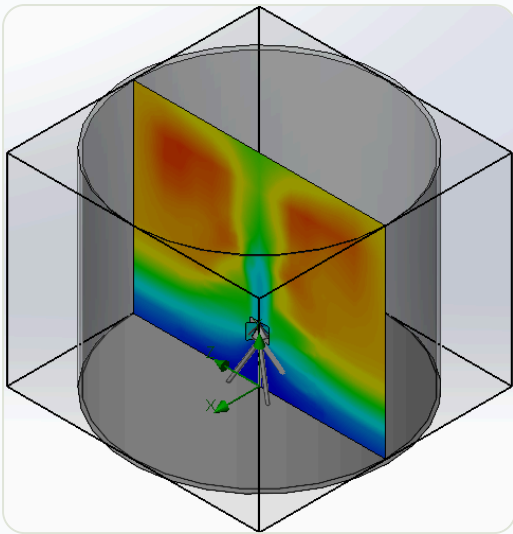
Full write-up in progress — imagery above. Industry: real estate / built-environment ventilation, Singapore.

CFD · MULTIPHASE

Roof Rainfall Project

Multiphase CFD study of rainfall on a roof, chaining three solvers to follow water from airborne droplet to standing pool.

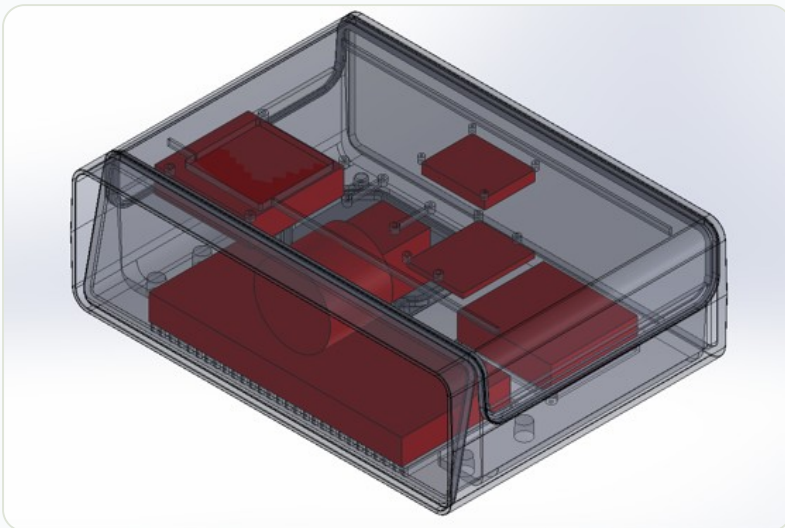
- Coupled Discrete Phase Model → Eulerian Wall Film → Volume of Fluid (DPM → EWF → VOF), so falling droplets, the thin film they form on the surface, and bulk pooling water were each handled by the appropriate physics model.
- Built as a feasibility test to prove the handoff between the three models could be made to work end-to-end.



CFD · MIXING

Industrial Mixing Tank — Thermal Study

Full write-up in progress — imagery above. Industry: industrial process engineering.



SIMULATION · SAFETY

Site Safety Monitoring Simulation

Full write-up in progress — imagery above. Industry: construction safety technology.



CFD · SPORTS AERO

KiteFoil Aerodynamics

Full write-up in progress — imagery above. Industry: sports equipment / marine aerodynamics.

PROCESS SIMULATION

Pad Production Line Simulation

Full write-up in progress — imagery above. Industry: consumer healthcare manufacturing.

Novartis

Scientific Modelling Intern — machine learning, simulation and AI tooling in pharmaceutical manufacturing.

ROLE	PERIOD	LOCATION	DOMAIN
Scientific Modelling Intern	Aug – Sept 2025	Basel, Switzerland	R&D · Pharma / ML

Over a summer internship in Basel I built machine-learning models, simulations and AI tools that removed manual work and cut costs across pharmaceutical R&D — spanning powder characterisation, process analytics and LLM-based automation.

APPROACH

1. Turn messy sensor and process data into clean, usable models.
2. Automate repetitive analysis with Python, dashboards and LLM workflows.
3. Validate against real batches and quantify the time and cost saved.

PROJECTS

5 projects

MACHINE LEARNING

ML Powder Flow Classification

Machine-learning models to predict powder flow characteristics from experimental data.

- Used LazyClassifier for rapid model selection, then Grid Search, Lasso regularisation and Bayesian optimisation for hyperparameter tuning.
- Saved CHF 40,000 per batch of APIs by predicting behaviour ahead of manufacture.

AI INFRASTRUCTURE

MCP Servers for Internal AI Tooling

Full write-up in progress.

DATA / PYTHON

Streamlit Analytics Dashboards

Interactive Streamlit applications that eliminated manual data searching for the department.

- Built dropdown-driven apps for dwell-time analysis and screw customisation.
- Developed Python programs for Loss-on-Drying calculations from sensor data, solving mass-balance equations for moisture content and drying efficiency.

LLMS · AGENTS

AI Project Summariser (LLMs + Agentic Workflows)

An AI summariser that gathers data from internal and external databases automatically.

- Used automated workflows and LLMs to compile project context on demand.
- Reduced research time from ~50 minutes to ~15, saving weeks of cumulative effort.

CAD · SIMULATION

Pill CAD Design for FEA / COMSOL

Tablet geometry modelled for high-performance-compute simulation.

- Designed tablets in SolidWorks from technical drawings for COMSOL HPC simulation.
- Saved thousands per batch through predictive modelling.

UCL

MEng Mechanical Engineering — the competition teams and projects that shaped how I engineer.

DEGREE	YEARS	PREDICTED	DOMAIN
MEng Mechanical Engineering	Sept 2024 – June 2028	First Class (83%)	Education

I'm in my second year of an MEng in Mechanical Engineering at UCL, predicted a First (83%). Alongside the degree I throw myself into engineering teams and competitions — rocketry, motorsport, design challenges and more — where the theory meets real hardware.

APPROACH

1. Apply core mechanical engineering: thermofluids, solid mechanics, dynamics, materials.
2. Specialise toward automotive and aerospace through team projects.
3. Lead and deliver — from budgets and Gantt charts to manufacture and test.

PROJECTS

4 projects

AEROSPACE · TEAM LEAD

UCL Racing — Rocket

National Rocketry Championship team lead, aerostructures & simulations engineer.

- Led an 8-engineer team with a £1,000 budget to 8th of 50 at the National Rocketry Championship.
- Validated ANSYS Fluent CFD against drag-coefficient data to 97% accuracy.
- Designed the aerostructure in Fusion 360, increasing apogee by 203 m.
- Ran the project in Jira with Gantt charts, risk assessments and sponsorship outreach.

AUTOMOTIVE · MOTORSPORT

Formula Student

Full write-up in progress.

DESIGN & MANUFACTURE

IMechE Design Challenge

Design & manufacturing engineer — runner-up, Greater London Regional Final.

- Secured National Final qualification as one of only two teams to achieve perfect accuracy.
- Won both the Design Award and the Challenge Competition.
- Used Fusion 360 Generative Design to cut mass by 115.87 g (60%) at a safety factor of 2.
- Produced a full Bill of Materials with data visualisation to identify cost drivers.

TURBINES · COMPETITION

UCL Mechathon 2025

Won the competition (team of 6) with a 76 mm-radius wind-turbine design.

- Achieved 29% efficiency at 3.3 V.
- Used QBlade to analyse aerofoil lift-to-drag ratios and optimal angle of attack (95.8% accurate).
- Developed manufacturing methods across milling, 3D printing and laser cutting.

PRESENTATIONS & COURSEWORK

Drop presentation PowerPoints and coursework PDFs into assets/docs/ and list them here.

Presentation / coursework 1

Presentation / coursework 2

Side Projects

The things I build for the love of it — research, robotics and everything between.

THEMES

Aerospace · Robotics · Research

TOOLS

CAD · Arduino · C++ · 3D print

STATUS

Ongoing

A selection of personal and academic projects I've taken end-to-end — from theoretical research papers to fully built, programmed hardware.

APPROACH

1. Start from a question or a problem worth solving.
2. Model, build and iterate — CAD, electronics, code and test.
3. Document the result properly, challenges and all.

PROJECTS

2 projects

RESEARCH · IMPACT

Internal Pressure of Thin-Walled Vessels During Collision

Theoretical research and mathematical modelling on pressure vs. impact force.

- Modelled the relationship between internal pressure and impact force.
- Wrote a 4,000-word research paper on optimal pressure for impact reduction and its applications to safety systems.

ROBOTICS · AUTOMATION

Automated Fruit & Vegetable Cutting Robot

A kitchen robot designed, built and programmed from scratch.

- Built and programmed the robot with Arduino and C++.
- Modelled and optimised components in CAD for functionality and durability.
- Assembled and iteratively refined the mechanical systems and control algorithms.